

Organic food consumption and the incidence of cancer in a large prospective study of women in the United Kingdom

Background: Organically produced foods are less likely than conventionally produced foods to contain pesticide residues. Methods: We examined the hypothesis that eating organic food may reduce the risk of soft tissue sarcoma, breast cancer, non-Hodgkin lymphoma and other common cancers in a large prospective study of 623 080 middle-aged UK women. Women reported their consumption of organic food and were followed for cancer incidence over the next 9.3 years. Cox regression models were used to estimate adjusted relative risks for cancer incidence by the reported frequency of consumption of organic foods. Results: At baseline, 30%, 63% and 7% of women reported never, sometimes, or usually/always eating organic food, respectively. Consumption of organic food was not associated with a reduction in the incidence of all cancer (n=53 769 cases in total) (RR for usually/always vs never=1.03, 95% confidence interval (CI): 0.99–1.07), soft tissue sarcoma (RR=1.37, 95% CI: 0.82–2.27), or breast cancer (RR=1.09, 95% CI: 1.02–1.15), but was associated for non-Hodgkin lymphoma (RR=0.79, 95% CI: 0.65–0.96). Conclusions: In this large prospective study there was little or no decrease in the incidence of cancer associated with consumption of organic food, except possibly for non-Hodgkin lymphoma.